12.5 Determining Tax, Tip, or Fees Given a Percentage

Lesson Introduction

Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.	Learning Targets	• I can determine the amount of tax, tip, or fee given a percentage. • I can determine the percentage of a discount or markup.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.	N/A

Essential Questions	• How do you determine the percentage of a discount or markup? • How do you calculate tax, tips, or fees for a given percentage?
Lesson Narrative	In this lesson, students extend their knowledge and skills with finding discounts and markups to determine the percentage used to discount or markup. Students also explore how to calculate taxes, tips, and fees in context.

Component Summary				
	12.5.1 Warm-Up (~5 min.)	Error analysis of sales flyer *(SE p. 235)*	12.5.4 Guided Instruction (10-15 min.)	Explore how to find taxes, tips, and fees *(SE p. 237)*
	12.5.2 Guided Instruction (10 min.)	Explore methods to determine the percentage of a discount or markup *(SE p. 235)*	12.5.5 Guided Practice (5 min.)	Practice determining tips, taxes, and fees *(SE p. 239)*
	12.5.3 Try It (5 min.)	Error analysis of calculating a percentage discount *(SE p. 237)*	12.5.6 Your Turn! (5 min.)	Practice finding taxes and percentages of discount and markup *(SE p. 240)*
	12.5.7 Wrap-Up (~5 min.)	Demonstrate ability to calculate the price of an order after tax *(SE p. 240)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms</u>: • Gratuity (Tip) • Markup • Discount • Markdown • Commission • Sales Tax • Subtotal • Total	• Calculators • (Optional) Chart paper • (Optional) Markers	• Consider which problems or components you may want students to solve without using a calculator. • Consider what calculations/work you want students to show in their workbook for problems where they use a calculator.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

- Students may struggle converting between different forms of rational numbers (i.e., fraction to decimal, percent to decimal, decimal to percent).
- Students may not understand the difference between the subtotal and the total and which is used when determining tax or tip amounts in various contexts.

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- During the 12.5.2 Guided Instruction, consider creating an anchor chart with instructions, examples, and visuals for finding percentage of a discount and markup.
- Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating discounts, markups, taxes, tips, and percentages. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers.
- There are multiple strategies for solving the problems in this lesson. Encourage students to use strategies that make sense for them as they build fluency in solving problems.

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<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
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Algebra Talk

In 12.5.1 Warm-Up, considering having students initially find and correct the error in the problem independently. Next, engage students in a discussion about the strategies students used to correct the error. This routine can also be used in debriefing 12.5.3 Try It.

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MA.K12.MTR.6.1: Reasonableness

In 12.5.6 Your Turn!, students determine the percentage of a discount and markup and find the tax in a real-world situation. Prompt students to informally check their answers by considering if the solution makes sense given the context. Then have students also formally check their calculations for accuracy.

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<tr><td>Differentiate Instruction</td><td colspan="2">In 12.5.2 and 12.5.4 Guided Instructions, components prompt students to annotate the examples using the Mark-Up strategy. Consider modeling this strategy at the conclusion of 12.5.4 Guided Instruction.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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12.5.1 Warm-Up: Math Fail

Component Overview: *Students identify and correct error(s) on a sale sticker. If students do not initially recognize the error(s), prompt them to calculate the discount and compare their work and answer to the discount on the sticker.*

12.5.1 Warm-Up: Math Fail

The following sticker was placed on an item for sale.

1. Describe any error(s) on the sticker.
   Answers will vary. Half price of $129 is not $99, it is $64.50. The sale price should be $64.50.

2. Correct the sticker so that it accurately reflects the "1/2 Price" advertised in the first line.
   $64.50

Algebra Talk
If time allows, consider using this language routine before moving on to the next component.
After students have completed the Warm-Up independently, ask students to engage in a discussion about the strategies they used to find and correct the error.

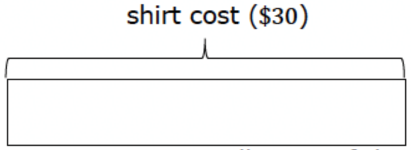
12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup

Component Overview: *Students are guided through a series of questions as they develop a stronger understanding of how to find discounts and markups in real-world scenarios. Students find the discount and markup amounts, final amounts, and the percentages of the discounts and markups.*

12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup

Recall that discount and markup problems typically include given percentages.

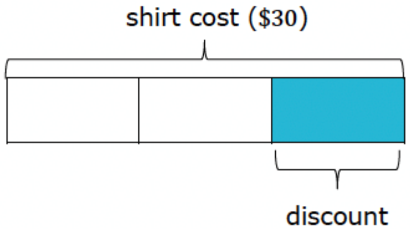
1. The tape diagram shows the original cost of a shirt, $\$30$.



A store manager wants to sell more of this shirt before the end of summer. The manager decides to put the shirt on sale for $1/3$ off the original price.

a. Will customers pay more or less than the original price for the shirt?
Less

The tape diagram is divided into thirds.



b. What is the price of the shirt after the discount?
$\$20$

c. How much will customers save from the original price?
$\$10$

d. Discuss with your partner how to calculate the percentage discounted if given the original price of the shirt and the sale price of the shirt.
$\frac{x}{100} = \frac{10}{30}; x = 33\%$
I used a percent proportion with the discounted amount to find what percentage 10 is of 30.

Consider creating an anchor chart with instructions, examples, and visuals for finding the percentage of a discount and markup.

Also, prompt students to underline keywords and annotate the examples in their own words. Debrief each component with a whole-class discussion to provide an auditory entry point.

- *How does the tape diagram show $\frac{1}{3}$ of the original price of the shirt?*
- *How do you know whether the customer will pay more or less than the original price from the question 1?*

Students could also use an equation to determine the percentage discounted.

$30x = 10$, *where x is the percentage discounted represented as a decimal.*

12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup (continued)

e. If the $\$30$ shirt was discounted to $\$18$, what percentage of the shirt was discounted.
   40%

2. Nova buys earrings from a supplier for $\$13.20$. She sells them in her boutique for $\$24.95$. Clare works at the boutique and was curious what percentage markup Nova uses. The table shows 2 different methods Clare used to determine the percentage the earrings are marked up when sold.

Method 1	Method 2
$\$24.95 = (1 + \text{Markup rate}) \times \$13.20$ $\frac{24.95}{13.20} = 1 +$ Markup rate $1.89 = 1 +$ Markup rate $1.89 - 1 =$ Markup rate	Nova paid $\$13.20$ Nova is selling for $\$24.95$ $\$24.95 - \$13.20 = \$11.75$ What % is 11.75 of 13.2? $\frac{11.75}{13.2} = 0.89$

a. Compare both methods. Discuss the similarities and differences in the two methods with your partner. Then, write them in the table.

Similarities	Differences
In both methods, Clare is trying to determine the markup rate. She finds the same markup rate using both strategies.	*Method 1 considers how the store computed the new sales price whereas method 2 finds the mark-up and determines what proportion of the original price this represents.*

b. What is the percentage of the markup?
   89%

Consider giving students the opportunity to reflect on whether they prefer one strategy to the other based on the methods shared by Clare.

12.5.3 Try It: Finding the Percentage of a Discount or Markup

Component Overview: *Students identify and correct error(s) in Johnetta's calculations of the discount percentage of a backpack.*

12.5.3 Try It: Finding the Percentage of a Discount or Markup

1. Johnetta calculated the percentage discounted for a backpack that normally costs $\$25$ but is on sale for $\$21$. Johnetta has made errors in her work. Identify Johnetta's errors and correct her work.

Backpack original cost is $\$25$

Backpack sale cost is $\$21$

$$\frac{\$25}{\$21} = 1.19$$

Discount 1.19%

Answers will vary. Johnetta should have found the amount of the discount, $\$4$, by subtracting 21 from 25. Then, she should have divided the difference by 25 and multiplied by 100 to get 16%.

$$\frac{\$4}{\$25} = 0.16$$

Discount 16%

She could have also set up a percent proportion or equation to solve.

$$\frac{(25-21)}{25} = \frac{x}{100}$$

$25x = 25 - 21$, *where* x *is the percentage discount represented as a decimal*

If students do not initially recognize the error(s), prompt them to calculate the discount percentage and compare their work and answers to Johnetta's.

Algebra Talk
Have students initially find and correct the error in the question independently. Next, engage students in a discussion about the strategies they used to correct the error.

12.5.4 Guided Instruction: Taxes, Tips, and Fees

Component Overview: *Students are guided through a series of questions as they develop an understanding of how to calculate taxes, tips, and fees in real-world scenarios. Explain the difference between the subtotal and the total to avoid student confusion.*

12.5.4 Guided Instruction: Taxes, Tips, and Fees

1. Baxter has been saving all of his money to purchase new Nike Air Max sneakers. After four months, he finally had $\$200$ saved, so he went to purchase the sneakers. Unfortunately, when Baxter checked out, the total came to $\$212$, which was $\$12$ more than he had saved.

Subtotal	\$200.00
Estimated Shipping	\$0.00
Estimated Tax	\$12.00
TOTAL	\$212.00

   a. Discuss with your partner why Baxter's total was more than \$200.
   The total is more than \$200 because he had to pay sales tax.

   b. If Baxter had to pay 8% sales tax on the same Nike Air Max sneakers, what would the cost of the taxes be?
   $200 \times 0.08 = \$16$

   c. Explain whether the estimated tax from Baxter's online search was more or less than 8%.
   The online sales tax was less because it was less than $\$16$. It was only 6% of the total.

Facilitate a brief discussion about how and when they have used or heard of taxes, tips, and fees in their everyday lives.

Work with your partner to complete the following.

2. The Cardone Family is planning to have a birthday celebration at one of their favorite restaurants. The restaurants they are considering are in different counties that each have different sales tax rates.

   a. Complete the table to determine how much tax will be at each restaurant given the same pre-tax cost of dinner. The first row of the table has been completed.

Restaurant	Cost of Dinner (Pre-tax)	How Much Sales Tax?
Restaurant A	$120	Sales Tax 7% $(120)(0.07) = 8.40$ $8.40 tax
Restaurant B	$120	Sales Tax 7.5% $9
Restaurant C	$120	Sales Tax 8% $9.60
Restaurant D	$120	Sales Tax 8.5% $10.20

   b. The Cardone Family plans to leave a tip for their server. They intend to tip 20% of the cost of their dinner. If the cost of dinner is $120, how much tip will the server receive?
   $24

3. Qualyen bought a new computer and printer online totaling $850. He had to pay an additional 3% in shipping fees on top of the $850. With your partner, determine the cost of the shipping fees.
   $25.50

- *What do you notice about the options for the Cardone family dinner?*
- *What are the similarities and differences in the restaurant choices?*
- *Based on the information provided, which restaurant is in the county with the largest tax rate? The smallest?*

Challenge students to create equations that can be used to calculate the tax, tip, and fee. Prompt students to check their equations and make any necessary revisions.

12.5.4 Guided Instruction: Taxes, Tips, and Fees (continued)

There are many situations where a percentage of an amount of money is added to or subtracted from a total amount. For example:

- If a restaurant bill is $\$34$ and the customer pays $\$40$, they left $\$6$ dollars as a tip for the server. That is 18% of $\$34$, so they left an 18% tip.
- If the price tag on a shirt in Arizona is $\$11.50$, then the sales tax is $(0.075) \cdot 11.5 = 0.8625$, which rounds to 86 cents. The customer pays $11.50 + 0.86$, or $\$12.36$ for the shirt.
- A student goes to a hairdresser to get ready for the 8^{th} grade formal dance. The total is $\$125$. The student's parents tip the hairdresser 15%. The total cost of the service is $\$143.75$.
- A gentleman purchases a suit originally priced at $\$397.98$. The suit is on clearance with a 45% off sticker. The discount reduces the suit $\$179.09$ off the original price, resulting in the new suit price being $\$218.89$.

	Goes to	How It Works
Sales Tax	The government	Added to the price of the item
Gratuity (tip)	The person providing the service	Added to the cost of the service
Markup	The seller	Added to the cost of an item so the seller can make a profit
Markdown (discount)	The customer	Subtracted from the price of an item to encourage the customer to buy it
Commission	The salesperson	Percentage of the value of the sale that is subtracted from the total sale and given to the salesperson

Encourage students to use the Mark-Up strategy as they read through the sample situations provided in this component.

Consider also structuring this with a Think Aloud for struggling readers where the teacher models how to mark up the text with annotations for future reference.

12.5.5 Guided Practice: Calculating Tax, Tip, and Fees

Component Overview: *Students practice determining tip, tax, and fees in real-world scenarios.*

12.5.5 Guided Practice: Calculating Tax, Tip, and Fees

1. Johnny went shopping for his mom at Save More Foods and his subtotal was $11.75. The tax applied at the grocery store is 6%. How much was the tax for his purchase?
   $0.71

2. Rosa and her friends are eating at a restaurant for dinner. Their bill was $45.80. They want to leave a 20% tip. How much tip will the server receive?
   $9.16

3. Rick bought 3 shirts for $18 each, 2 pairs of socks for $3.99 a pair, and a pair of slacks for $45.00. The sales tax rate is 8.5%. How much sales tax did Rick pay?
   $9.09

How do you rewrite 6% as a decimal?

Use mental math to determine 10% of $45.80. How can you use this value to determine the tip the server will receive in question 2?

12.5.6 Your Turn!

Component Overview: *Students are provided with additional practice calculating sales tax and the discount and markup percentages.*

12.5.6 Your Turn!

1. A city in Ohio has a sales tax rate of 7.25%. Complete the table to determine how much sales tax is for each item.

Item	Price Before Tax ($)	Sales Tax ($)
Pillow	8.69	$0.63
Blanket	22.00	$1.60
Trash Can	14.50	$1.05

2. A store has a sale on pants. The original price of the pants is $19.00. With the discount, the price of one pair of pants before tax is $15.20. What was the percent discount?
   20%

3. A music store marks up its instruments. The store bought a guitar for $45 and sold it for $58.50. The store bought a trumpet for $80 and sold it for $104. What is the percent markup the store used to price the guitar and trumpet?
   30%

Consider asking students the following:

- *How did you calculate the sales tax?*
- *Why did you select this method?*
- *Will this method always work?*

12.5.7 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their ability to calculate the amount of sales tax applied to an order. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should be allowed to access a calculator to complete the Wrap-Up.*

12.5.7 Wrap-Up: Ticket Out the Door

1. Mrs. Jones attended a Southern Living home party. She purchased a vase that cost $\$34.99$ plus $\$3.00$ for shipping. There is a 8.25% sales tax on the cost of the vase and shipping charge.

   What is the amount of tax Mrs. Jones paid on her purchase?
   $\$3.13$

Remind students that they can reference similar problems they have solved in the lesson. Prompt students to check the reasonableness of their answers.